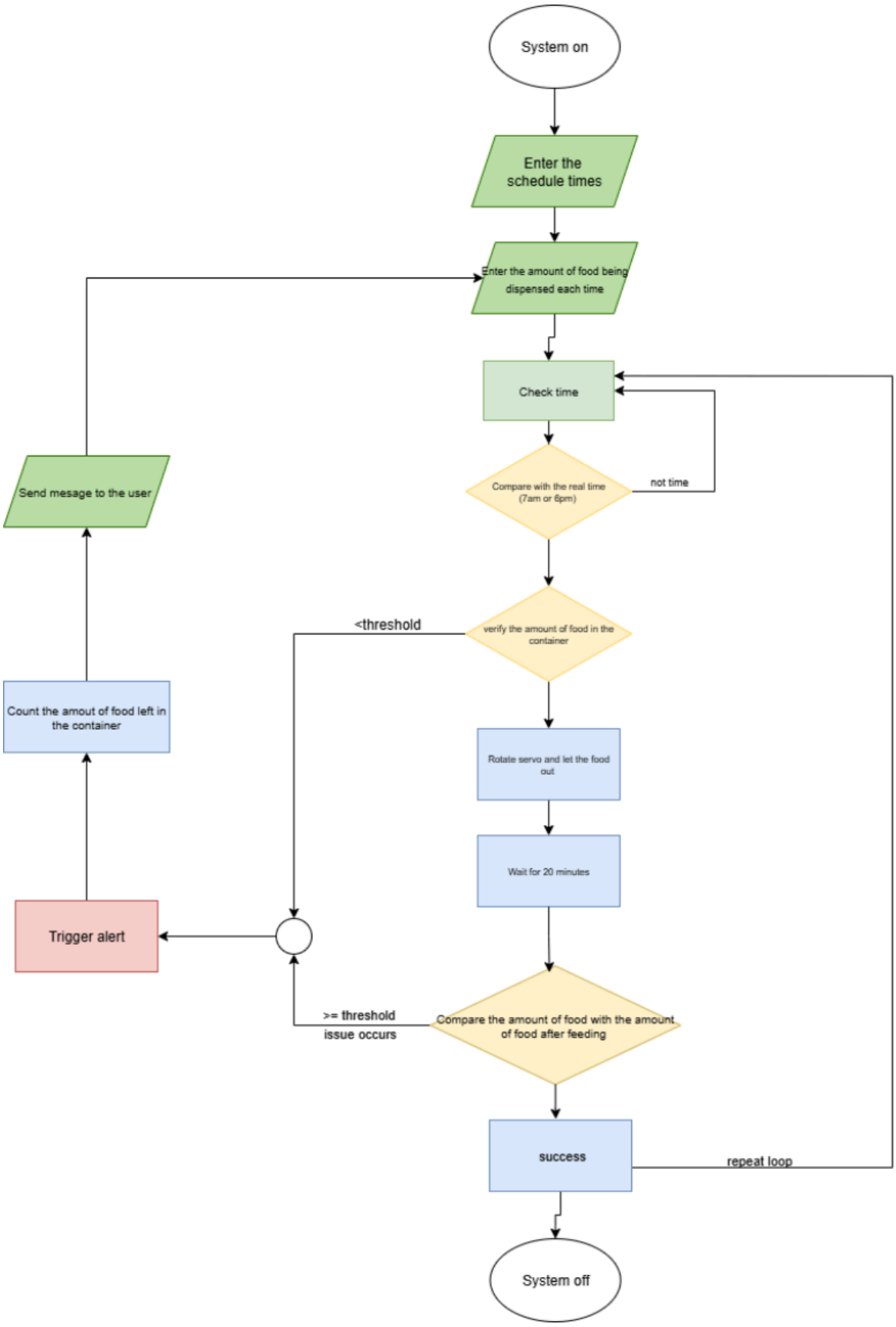
Step 3: Plan the Solution (Design the Algorithm)

In this phase, the algorithm is made to control the operation of the automated pet feeder. Its main tasks are to create clear decision logic to identify when and how the system should dispense food and send alerts to any issues, such as an empty food container or uneaten food, which are to be promptly solved. To ensure the system functions are well-designed before implementation and further testing, the flowchart below visually represents the algorithm.


System on
Enter the schedule times
Enter the amount of food being dispensed each time
Check time
Compare with the real time (7am or 6pm)
not time
verify the amount of food in the container
<threshold
Rotate servo and let the food out
Wait for 20 minutes
Compare the amount of food with the amount of food after feeding
>= threshold
issue occurs
success
repeat loop
System off
Trigger alert
Count the amout of food left in the container
Send mesage to the user